

Optimal Control of the Ethena Yield-Bearing Stablecoin

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This version: May 11, 2026

Abstract

We formulate and solve stochastic control problems that model the core yield-generating strategy of the Ethena protocol, a decentralized finance (DeFi) stablecoin that earns yield by combining a long position in staked Ethereum (stETH) with an equal-sized short position in ETH perpetual futures. The combined position is delta-neutral with respect to the ETH spot price, yet earns carry from two sources: staking rewards on the stETH leg, and funding-rate payments received from long perpetual holders when the perpetual trades at a premium to spot. A key feature of our model is that the control — the rate of simultaneously buying stETH and shorting the perpetual — exerts two distinct types of price impact. *Permanent* impact shifts the mid-market prices of both legs, compressing the basis and permanently eroding future funding income; this is modeled with separate spot and perpetual permanent-impact coefficients μ_1 and μ_2 . *Temporary* impact reflects execution slippage on each leg, modeled with separate coefficients λ_1 and λ_2 ; it enters only as a direct quadratic execution cost $\lambda \gamma_t^2$ with $\lambda = \lambda_1 + \lambda_2$. We study both an infinite-horizon discounted problem and a finite-horizon problem in which the protocol maximizes total wealth up to a fixed date T , subject to a terminal cost for liquidating any remaining position. In both cases the optimal control is obtained explicitly: in the infinite-horizon case via a quadratic ansatz for the Hamilton–Jacobi–Bellman (HJB) equation, and in the finite-horizon case via a time-dependent quadratic ansatz that reduces the HJB partial differential equation to a system of Riccati ordinary differential equations. In both settings the optimal control is linear in the state (D_t^Y, N_t^Y) : the protocol should build its position faster when the basis is wide and taper accumulation as the position grows and inventory risk accumulates. Near the terminal date the finite-horizon policy additionally generates a liquidation urgency that is absent from the infinite-horizon solution.

Keywords: stochastic control, Hamilton–Jacobi–Bellman equation, perpetual futures, funding rate, staking yield, price impact, DeFi, stablecoin, Ornstein–Uhlenbeck process.

1 INTRODUCTION

Decentralized finance (DeFi) has given rise to a new class of yield-bearing stablecoins that seek to combine price stability with attractive returns through explicitly structured financial strategies. Among the most prominent of these is the Ethena protocol ([Ethena Labs, 2024](#)), which issues a dollar-denominated stablecoin whose yield is generated by a *delta-neutral* strategy: deposits of USD-denominated collateral are used to purchase staked Ether (stETH), while an equal notional short position in ETH perpetual futures is simultaneously established. Because the long stETH position and the short perpetual position respond identically to movements in the ETH spot price, the combined book

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is market-neutral in ETH, yet earns two streams of carry. First, the stETH earns a continuous staking yield. Second, because the ETH perpetual futures price typically trades at a premium to the spot price, the funding mechanism of perpetual contracts requires long holders to make periodic cash payments to short holders; as the short leg of the trade, Ethena receives these payments.

The Ethena protocol has grown remarkably quickly since USDe launched in February 2024 ([Messari, 2024](#)). Within four months of launch, USDe surpassed a \$3 billion market capitalization to become the fourth-largest stablecoin, a milestone its founder claimed had been reached faster than any other USD-denominated crypto asset in history ([Binance, 2024](#)). Growth accelerated further in 2025: by August of that year USDe had climbed to a market capitalization of approximately \$9.3 billion, briefly ranking it as the third-largest stablecoin behind only Tether’s USDT and Circle’s USDC ([The Block, 2025](#)), before reaching a peak of nearly \$15 billion later in the year ([CoinGecko, 2026a](#)). As of the time of writing (May 2026), USDe carries a market capitalization of approximately \$4 billion, ranking it among the thirtieth-largest cryptocurrencies overall and making it the largest stablecoin that is not backed by government bonds or fiat bank deposits ([CoinGecko, 2026b](#); [CoinMarketCap, 2026](#)). The protocol’s rapid ascent has attracted considerable attention from both practitioners and academics, and raises fundamental questions about the optimal dynamic management of the delta-neutral strategy that underlies it.

The funding-rate mechanism in perpetual futures markets is designed to keep the perpetual price anchored to the underlying spot. When the perpetual trades at a premium (positive basis), longs pay shorts at a rate proportional to the size of the premium; see [BitMEX \(2016\)](#). Empirically, ETH perpetual markets have spent the majority of historical time in positive-funding environments, reflecting persistent net demand for leveraged long exposure ([Deribit Insights, 2022](#)), making the short perpetual overlay structurally advantageous.

While the strategy is conceptually simple, its *optimal dynamic implementation* is non-trivial and involves several interacting forces. When the protocol grows its position — buying stETH on the spot market and shorting the perpetual — it exerts two distinct types of price impact. First, there is *permanent* impact: buying stETH shifts the spot mid-price upward, and shorting the perpetual shifts the perpetual mid-price downward; both effects compress the perpetual-spot basis and permanently erode the funding premium from which the protocol earns its carry. Second, there is *temporary* impact: market orders on each leg fill at prices worse than the mid, generating immediate execution slippage. We model these two channels with entirely separate parameters, and show that both must be controlled in an optimal implementation. At the same time, a large outstanding position creates inventory risk: in episodes of negative funding (when the basis inverts and longs receive payments from shorts), the protocol incurs carry costs proportional to its short position size.

The literature on optimal execution and inventory management provides the natural framework for these trade-offs. [Almgren and Chriss \(2001\)](#) pioneered the use of quadratic execution costs to model market impact in optimal liquidation, yielding tractable linear-quadratic stochastic control problems with explicit solutions. [Cartea et al. \(2015\)](#) and [Guéant \(2016\)](#) provide comprehensive treatments of optimal trading with stochastic price dynamics and inventory penalties. Mean-reverting spread processes similar to our basis model appear in pairs trading and statistical arbitrage; see [Jurek and Yang \(2007\)](#) and [Liu and Longstaff \(2005\)](#). Stochastic control in DeFi has received growing attention: [Cartea et al. \(2023\)](#) study optimal liquidity provision in automated market makers, and perpetual futures mechanics are analyzed in [Ackerer et al. \(2024\)](#).

Our contributions are as follows. We formulate continuous-time stochastic control problems capturing the Ethena trade-offs: permanent two-sided price impact from simultaneous spot and perpetual order flow (which feeds back into

the basis dynamics), separate temporary execution costs from market-order slippage on each leg, carry income from staking and funding, and a quadratic inventory risk penalty. We treat both an infinite-horizon discounted formulation and a finite-horizon formulation in which the protocol maximizes total accumulated wealth up to a fixed liquidation date. In both cases we obtain the value function and optimal control explicitly. In the infinite-horizon case the solution follows from a quadratic ansatz for the stationary HJB equation, reducing the problem to a cascade of scalar algebraic equations. In the finite-horizon case the same quadratic ansatz reduces the HJB PDE to a backward system of Riccati ODEs, which inherits the same hierarchical structure and admits an explicit integral formula for the affine (non-quadratic) coefficients. In both settings the optimal control is a linear feedback in the current basis and position size; near the terminal date the finite-horizon feedback coefficients generate a liquidation urgency absent from the infinite-horizon solution.

The rest of this paper is organized as follows. Section 2 presents the model. Section 3 formulates and solves an infinite-horizon stochastic control problem. Section 4 formulates and solves a finite-horizon stochastic control problem. Some numerical experiments are performed in Section 5. Section 6 offers some concluding remarks and possible directions for future research.

2 MODEL

2.1 POSITIONS, PRICES, AND EXECUTION COSTS

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, and let $W = \{W_t\}_{t \geq 0}$ be a standard Brownian motion. The Ethena protocol simultaneously holds a long position of N_t^γ units of staked ETH and a short position of N_t^γ units of the ETH perpetual futures contract. The position size $N_t^\gamma \in \mathbb{R}$ can be positive (the intended long-stETH, short-perp carry trade) or negative (a reversed position); as we show in Section 3, the optimal policy generically keeps $N^\gamma > 0$. The control variable is $\gamma_t = dN_t^\gamma/dt$: the rate of buying stETH and simultaneously shorting the perpetual when $\gamma_t > 0$, or unwinding both legs when $\gamma_t < 0$. Here and throughout this paper, stochastic processes that depend on γ will carry it as a superscript.

Let S_t^γ denote the mid-market spot price of ETH and P_t^γ the mid-market price of the ETH perpetual futures, so that the *basis* $D_t^\gamma := P_t^\gamma - S_t^\gamma$ is the perpetual premium over spot. When $D^\gamma > 0$, the perpetual trades at a premium and the funding mechanism pays longs to shorts; the Ethena protocol, being short the perpetual, receives these payments.

When the Ethena protocol trades at rate γ_t , it faces two-sided *temporary* execution costs reflecting the fact that market orders fill at prices worse than the mid. On the spot leg, buying stETH at rate γ_t moves the execution price above the mid by $\lambda_1 \gamma_t$, so the protocol pays $S_t^\gamma + \lambda_1 \gamma_t$ per unit purchased, where $\lambda_1 \geq 0$ is the spot temporary impact coefficient. On the perpetual leg, shorting at rate γ_t moves the execution price below the mid by $\lambda_2 \gamma_t$, so the protocol receives $P_t^\gamma - \lambda_2 \gamma_t$ per unit shorted, where $\lambda_2 \geq 0$ is the perpetual temporary impact coefficient. The overpayment on the spot leg is $\lambda_1 \gamma_t$ per unit times γ_t units per unit time, contributing a cost of $\lambda_1 \gamma_t^2$ per unit time; similarly the perpetual leg contributes $\lambda_2 \gamma_t^2$. The total instantaneous temporary execution cost is therefore

$$\text{temporary execution cost rate} = (\lambda_1 + \lambda_2) \gamma_t^2 =: \lambda \gamma_t^2, \quad (1)$$

where we have defined $\lambda := \lambda_1 + \lambda_2 > 0$.

2.2 BASIS DYNAMICS AND PERMANENT PRICE IMPACT

Separately from the temporary execution costs above, trading at rate γ_t also has a *permanent* effect on mid-market prices that persists after the trade is complete. Buying stETH at rate γ_t shifts the spot mid S^γ upward permanently at rate $\mu_1 \gamma_t$, and shorting the perpetual at rate γ_t shifts the perpetual mid P^γ downward permanently at rate $\mu_2 \gamma_t$, where $\mu_1, \mu_2 \geq 0$ are the permanent impact coefficients (distinct from the temporary impact coefficients λ_1, λ_2). As $D^\gamma = P^\gamma - S^\gamma$, both permanent shifts act in the same direction to compress the basis:

$$\text{permanent impact on } D^\gamma : \quad -(\mu_1 + \mu_2) \gamma_t dt =: -\mu \gamma_t dt, \quad \mu := \mu_1 + \mu_2.$$

Absent order flow, the basis mean-reverts to a long-run level $m > 0$ at speed $\kappa > 0$ and is subject to random shocks. Combining the natural mean-reversion, the permanent impact, and the noise, the controlled basis dynamics are

$$dD_t^\gamma = \left[-\kappa (D_t^\gamma - m) - \mu \gamma_t \right] dt + c dW_t, \quad (2)$$

where $c > 0$ is the basis volatility. Because the impact is permanent, the shift $-\mu \gamma_t dt$ alters D_t^γ for all future times: the protocol must internalize that aggressive accumulation today permanently erodes the funding premium it will earn tomorrow. The position size evolves as

$$dN_t^\gamma = \gamma_t dt, \quad (3)$$

The state of the system is the pair $(D_t^\gamma, N_t^\gamma) \in \mathbb{R}^2$.

2.3 NET WEALTH DYNAMICS

Let X_t^γ denote the net wealth of the Ethena protocol. Changes in the value of X_t^γ over the period $[t, t + dt]$ arise from three sources.

(i) *Carry income*. The long stETH position earns staking income at constant rate $r > 0$ per unit, contributing $r N_t^\gamma dt$. The short perpetual position receives funding payments at rate $q D_t^\gamma$ per unit (positive when $D > 0$, i.e. when longs pay shorts), contributing $q D_t^\gamma N_t^\gamma dt$.

(ii) *Mark-to-market P&L*. The long stETH leg has mark-to-market (MTM) gain $N_t^\gamma dS_t^\gamma$ and the short perpetual leg has MTM gain $-N_t^\gamma dP_t^\gamma$. Their sum is $N_t^\gamma (dS_t^\gamma - dP_t^\gamma) = -N_t^\gamma dD_t^\gamma$: the dS_t^γ exposures cancel exactly (delta-neutrality), leaving only basis risk. Substituting (2), we have

$$-N_t^\gamma dD_t^\gamma = \left[\kappa (D_t^\gamma - m) + \mu \gamma_t \right] N_t^\gamma dt - c N_t^\gamma dW_t.$$

The term $\mu \gamma_t N_t^\gamma dt$ is the immediate MTM gain from the protocol's own permanent impact: the spot mid rises by $\mu_1 \gamma_t dt$ (gaining $\mu_1 \gamma_t N_t^\gamma$ on the long stETH leg) and the perpetual mid falls by $\mu_2 \gamma_t dt$ (gaining $\mu_2 \gamma_t N_t^\gamma$ on the short perp leg), summing to $\mu \gamma_t N_t^\gamma dt$. Note that this gain is not “free”: it is offset in expectation by the permanently lower future funding income caused by the compressed basis.

(iii) *Execution cost*. The total execution cost derived in (1) is $\lambda \gamma_t^2 dt$.

Combining (i), (ii) and (iii), we have

$$\begin{aligned} dX_t^\gamma &= r N_t^\gamma dt + q D_t^\gamma N_t^\gamma dt - N_t^\gamma dD_t^\gamma - \lambda \gamma_t^2 dt \\ &= \left[(q + \kappa) D_t^\gamma - \kappa m + r + \mu \gamma_t \right] N_t^\gamma dt - \lambda \gamma_t^2 dt - c N_t^\gamma dW_t. \end{aligned} \quad (4)$$

REMARK 1 (PERMANENT VS. TEMPORARY IMPACT). The model contains two distinct impact channels with separate parameters. The *permanent* impact coefficients μ_1 (spot) and μ_2 (perp) govern how much the mid-market prices shift lastingly per unit of order flow; their sum $\mu = \mu_1 + \mu_2$ enters the basis SDE (2) and generates the MTM gain $\mu \gamma_t N_t^\gamma dt$ in (4). The *temporary* impact coefficients λ_1 (spot) and λ_2 (perp) determine the execution-price slippage $\lambda = \lambda_1 + \lambda_2$ that appears only as the direct cost $\lambda \gamma_t^2 dt$ in the objective. The two sets of parameters are independent: μ and λ can take any non-negative values.

3 INFINITE-HORIZON STOCHASTIC CONTROL PROBLEM

3.1 VALUE FUNCTION AND HJB EQUATION

We introduce a quadratic inventory risk penalty $\varphi \cdot (N_t^\gamma)^2$ (with $\varphi > 0$) to penalize large positions, capturing basis variance risk (the term $-cN_t^\gamma dW_t$ in (4) has variance $c^2 \cdot (N_t^\gamma)^2 dt$), liquidation risk, and the cost of sustained negative funding. Let $\mathcal{G}$ denote the class of admissible controls consisting of progressively measurable processes $\gamma = \{\gamma_t\}_{t \geq 0}$ for which (2)–(3) admit a unique strong solution. We define the infinite-horizon value function as follows

$$\begin{aligned} V(d, n) &:= \sup_{\gamma \in \mathcal{G}} \mathbb{E} \left[\int_0^\infty e^{-\rho t} \left(dX_t^\gamma - \varphi \cdot (N_t^\gamma)^2 dt \right) \mid D_0^\gamma = d, N_0^\gamma = n \right] \\ &= \sup_{\gamma \in \mathcal{G}} \mathbb{E} \left[\int_0^\infty e^{-\rho t} \left([(q + \kappa)D_t^\gamma - \kappa m + r + \mu \gamma_t] N_t^\gamma - \lambda \gamma_t^2 - \varphi \cdot (N_t^\gamma)^2 \right) dt \mid D_0^\gamma = d, N_0^\gamma = n \right], \end{aligned} \quad (5)$$

where $\rho > 0$ is the discount rate. The three terms in the integrand represent, respectively: the expected rate of wealth increase from carry income and permanent-impact MTM; the total execution cost; and the inventory risk penalty.

By the principle of dynamic programming, the value function $V : \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfies the HJB equation

$$\rho V(d, n) = \sup_{\gamma \in \mathbb{R}} \left\{ [(q + \kappa)d - \kappa m + r + \mu \gamma] n - \lambda \gamma^2 - \varphi n^2 + [-\kappa(d - m) - \mu \gamma] \partial_d V + \gamma \partial_n V + \frac{c^2}{2} \partial_{dd} V \right\}. \quad (6)$$

The supremum over γ is attained at the unconstrained maximizer obtained from the first-order condition

$$\mu n - 2\lambda \gamma - \mu \partial_d V + \partial_n V = 0, \quad (7)$$

giving the candidate optimal feedback control

$$\gamma^*(d, n) = \frac{\mu n - \mu \partial_d V(d, n) + \partial_n V(d, n)}{2\lambda}. \quad (8)$$

The term μn in the numerator of (8) — absent in standard execution problems with only temporary impact — arises from the permanent impact MTM benefit $\mu \gamma N^\gamma$ in the running reward: a larger existing position makes it more attractive to keep accumulating, since each new unit sold short generates an immediate MTM gain of $\mu \gamma dt$ per existing unit. Substituting (8) into (6) gives the reduced HJB:

$$\rho V = [(q + \kappa)d - \kappa m + r] n - \varphi n^2 - \kappa(d - m) \partial_d V + \frac{(\mu n - \mu \partial_d V + \partial_n V)^2}{4\lambda} + \frac{c^2}{2} \partial_{dd} V. \quad (9)$$

3.2 EXPLICIT SOLUTION

The linear-quadratic structure of (9) motivates the ansatz

$$V(d, n) = \alpha_1 n^2 + \alpha_2 n d + \alpha_3 n + \alpha_4 d^2 + \alpha_5 d + \alpha_6, \quad (10)$$

with coefficients $\alpha_1, \dots, \alpha_6 \in \mathbb{R}$ to be determined. The partial derivatives are

$$\partial_d V = \alpha_2 n + 2\alpha_4 d + \alpha_5, \quad \partial_n V = 2\alpha_1 n + \alpha_2 d + \alpha_3, \quad \partial_{dd} V = 2\alpha_4. \quad (11)$$

Define the combined quantity

$$\begin{aligned} \Phi(d, n) &:= \mu n - \mu \partial_d V + \partial_n V \\ &= (\mu(1 - \alpha_2) + 2\alpha_1)n + (\alpha_2 - 2\mu\alpha_4)d + (\alpha_3 - \mu\alpha_5). \end{aligned}$$

If the value function is of the form (10), we can express the optimal control as $\gamma^* = \Phi/(2\lambda) = \gamma_N n + \gamma_D d + \gamma_0$, where

$$\gamma_N := \frac{\mu(1 - \alpha_2) + 2\alpha_1}{2\lambda}, \quad \gamma_D := \frac{\alpha_2 - 2\mu\alpha_4}{2\lambda}, \quad \gamma_0 := \frac{\alpha_3 - \mu\alpha_5}{2\lambda}. \quad (12)$$

We are now in a position to present our main result for the infinite-horizon stochastic control problem.

PROPOSITION 2 (EXPLICIT SOLUTION). *Suppose $\rho > 0$, $\kappa > 0$, $\varphi > 0$, $\mu = \mu_1 + \mu_2 \geq 0$, $\lambda = \lambda_1 + \lambda_2 > 0$, $r > 0$, $q > 0$, and the stability condition*

$$\kappa_* := \rho + 2\kappa - c^2 > 0. \quad (13)$$

Then the value function (5) is given by the quadratic (10) and the optimal control is the linear feedback

$$\gamma^*(D_t^\gamma, N_t^\gamma) = \gamma_N N_t^\gamma + \gamma_D D_t^\gamma + \gamma_0, \quad (14)$$

where the feedback coefficients γ_N , γ_D , γ_0 are given by (12), the coefficients α_1 and α_4 are the admissible roots of the quadratics (15) and (17), given explicitly by (16) and (18), the coefficient α_2 solves the fixed-point equation (19), the coefficients α_3 and α_5 solve the linear system (22)–(23) with explicit solution (25), and α_6 is given by (26).

Proof. Substitute (10)–(11) into the reduced HJB (9), expand Φ^2 , and collect by monomial in (d, n) .

Coefficient of n^2 :

$$\rho\alpha_1 = -\varphi + \lambda\gamma_N^2, \quad \text{i.e.,} \quad 4\lambda\rho\alpha_1 = -4\lambda\varphi + [\mu(1 - \alpha_2) + 2\alpha_1]^2.$$

Expanding and collecting terms in α_1 :

$$4\alpha_1^2 + [4\mu(1 - \alpha_2) - 4\lambda\rho]\alpha_1 + \mu^2(1 - \alpha_2)^2 - 4\lambda\varphi = 0. \quad (15)$$

The admissible root (giving $\alpha_1 < 0$ so that V is concave in n) is

$$\alpha_1 = \frac{[\lambda\rho - \mu(1 - \alpha_2)] - \sqrt{[\lambda\rho - \mu(1 - \alpha_2)]^2 + 4\lambda\varphi - \mu^2(1 - \alpha_2)^2}}{2}. \quad (16)$$

Coefficient of d^2 :

$$\rho\alpha_4 = \lambda\gamma_D^2 - 2\kappa\alpha_4 + c^2\alpha_4, \quad \text{i.e.,} \quad 4\lambda\kappa_*\alpha_4 = (\alpha_2 - 2\mu\alpha_4)^2,$$

where $\kappa_* = \rho + 2\kappa - c^2 > 0$. This yields the quadratic

$$4\mu^2\alpha_4^2 - (4\mu\alpha_2 + 4\lambda\kappa_*)\alpha_4 + \alpha_2^2 = 0, \quad (17)$$

with admissible root

$$\alpha_4 = \frac{(\mu\alpha_2 + \lambda\kappa_*) - \sqrt{(\mu\alpha_2 + \lambda\kappa_*)^2 - \mu^2\alpha_2^2}}{2\mu^2} = \frac{(\mu\alpha_2 + \lambda\kappa_*) - \sqrt{\lambda\kappa_*(\lambda\kappa_* + 2\mu\alpha_2)}}{2\mu^2}. \quad (18)$$

Coefficient of nd :

$$(\rho + \kappa)\alpha_2 = (q + \kappa) + 2\lambda\gamma_N\gamma_D = (q + \kappa) + \frac{[\mu(1 - \alpha_2) + 2\alpha_1](\alpha_2 - 2\mu\alpha_4)}{2\lambda}. \quad (19)$$

Since α_1 and α_4 depend on α_2 through (16)–(18), equation (19) is a scalar nonlinear fixed-point equation for α_2 . It can be solved by iteration starting from $\alpha_2^{(0)} = (q + \kappa)/(\rho + \kappa)$, which is the solution when the coupling terms on the right-hand side are negligible.

Coefficients of n and d (linear terms): Collecting the coefficient of n (constant in d), using $-\kappa(d - m)\partial_d V \ni \kappa m\alpha_2 n$ and the cross term $\Phi^2/(4\lambda) \ni \gamma_N\gamma_0 n$:

$$\rho\alpha_3 = r + \kappa m\alpha_2 + \frac{[\mu(1 - \alpha_2) + 2\alpha_1](\alpha_3 - \mu\alpha_5)}{2\lambda} = r + \kappa m\alpha_2 + \gamma_N(\alpha_3 - \mu\alpha_5). \quad (20)$$

Collecting the coefficient of d (constant in n), using $-\kappa(d - m)\partial_d V \ni -\kappa\alpha_5 d + 2\kappa m\alpha_4 d$ and the cross term $\ni \gamma_D\gamma_0 d$:

$$(\rho + \kappa)\alpha_5 = 2\kappa m\alpha_4 + \frac{(\alpha_2 - 2\mu\alpha_4)(\alpha_3 - \mu\alpha_5)}{2\lambda} = 2\kappa m\alpha_4 + \gamma_D(\alpha_3 - \mu\alpha_5). \quad (21)$$

Writing $C := \alpha_3 - \mu\alpha_5$, equations (20)–(21) become

$$\rho\alpha_3 - \gamma_N C = r + \kappa m\alpha_2, \quad (22)$$

$$(\rho + \kappa)\alpha_5 - \gamma_D C = 2\kappa m\alpha_4. \quad (23)$$

From the definition $C = \alpha_3 - \mu\alpha_5$, multiply (23) by μ and subtract from (22):

$$\rho\alpha_3 - \mu(\rho + \kappa)\alpha_5 - (\gamma_N - \mu\gamma_D)C = r + \kappa m\alpha_2 - 2\mu\kappa m\alpha_4. \quad (24)$$

Substituting $\alpha_3 = C + \mu\alpha_5$ into (24) yields a linear equation for C alone:

$$[\rho - \gamma_N + \mu\gamma_D - \mu(\rho + \kappa) + \rho\mu]C + \mu\rho C = r + \kappa m\alpha_2 - 2\mu\kappa m\alpha_4 + \mu(\rho + \kappa) \cdot 0,$$

which simplifies to

$$[\rho - \gamma_N + \mu\gamma_D]C = r + \kappa m(\alpha_2 - 2\mu\alpha_4).$$

Hence

$$C = \alpha_3 - \mu\alpha_5 = \frac{r + \kappa m(\alpha_2 - 2\mu\alpha_4)}{\rho - \gamma_N + \mu\gamma_D} = \frac{r + 2\kappa m\lambda\gamma_D}{\rho - \gamma_N + \mu\gamma_D}, \quad (25)$$

provided the denominator is nonzero (guaranteed under (13)). Substituting back into (22) gives $\alpha_3 = (r + \kappa m\alpha_2 + \gamma_N C)/\rho$, and then $\alpha_5 = (\alpha_3 - C)/\mu$.

Constant term:

$$\rho\alpha_6 = \lambda\gamma_0^2 + c^2\alpha_4 + \kappa m\alpha_5. \quad (26)$$

This completes the determination of all six coefficients. $\square$

REMARK 3 (ROOT SELECTION AND CONCAVITY). The minus sign before the square root in (16) yields $\alpha_1 < 0$, ensuring that V is strictly concave in n and the supremum in (5) is finite. Concavity in n requires $4\lambda\varphi > \mu^2(1 - \alpha_2)^2$; when φ is large relative to μ^2/λ this is readily satisfied. The root (18) with the minus sign gives $\alpha_4 \geq 0$: the value function is convex in d , reflecting the fact that a wider basis is unambiguously beneficial (more funding income). Standard verification arguments (Fleming and Soner, 2006) confirm the candidate solution equals the true value function.

REMARK 4 (CLOSED-LOOP DYNAMICS). Under the optimal control (14), the joint dynamics of (D, N) are

$$dD_t^\gamma = [-(\kappa + \mu\gamma_D)D_t^\gamma - \mu\gamma_N N_t^\gamma + \kappa m - \mu\gamma_0]dt + c dW_t, \quad (27)$$

$$dN_t^\gamma = [\gamma_N N_t^\gamma + \gamma_D D_t^\gamma + \gamma_0]dt. \quad (28)$$

The effective mean-reversion speed of the basis under the optimal policy is $\kappa + \mu\gamma_D \geq \kappa$, which is *faster* than the exogenous mean-reversion κ : the protocol's accumulation further compresses the basis, supplementing the natural reversion. Stability of the two-dimensional affine system requires the drift matrix

$$M := \begin{pmatrix} -(\kappa + \mu\gamma_D) & -\mu\gamma_N \\ \gamma_D & \gamma_N \end{pmatrix} \quad (29)$$

to have eigenvalues with strictly negative real parts, which is guaranteed by the conditions of Proposition 2. The stationary distribution of (D^γ, N^γ) under (27)–(28) is Gaussian and concentrates around $N^\gamma > 0$ whenever $m > 0$ and $r > 0$.

3.3 ECONOMIC IMPLICATIONS

The closed-form solution of Proposition 2 reveals several economically important features of the optimal Ethena strategy.

FUNDING-RATE CHASING. The feedback coefficient $\gamma_D = (\alpha_2 - 2\mu\alpha_4)/(2\lambda)$ governs how aggressively the protocol responds to the current basis D_t^γ . A wider basis raises both the instantaneous funding income $(q + \kappa)D \cdot N$ and the mark-to-market benefit of accumulation $(\mu\gamma_N)$, so we expect $\gamma_D > 0$: the protocol should build its position faster when the perpetual premium is high. Conversely, when the basis is narrow or negative, the optimal policy reduces γ_t (slows accumulation or unwinds), since funding income is low and the risk of further basis compression is not worth bearing.

TWO-SIDED PERMANENT IMPACT AND BASIS SELF-COMPRESSION. A central novelty of our model is the two-sided permanent impact: buying stETH shifts the spot mid upward at rate $\mu_1\gamma_t$ and shorting the perpetual shifts the perpetual mid downward at rate $\mu_2\gamma_t$, jointly compressing the basis through $\mu = \mu_1 + \mu_2$. The closed-loop basis dynamics (27) reveal that the effective mean-reversion speed becomes $\kappa + \mu\gamma_D$, which exceeds the exogenous κ : the protocol's own accumulation supplements natural reversion, limiting the duration of high-funding environments. The optimal policy internalizes this self-compression and moderates γ_D as μ grows. The individual values of μ_1 and μ_2 do not affect the optimal control (only $\mu = \mu_1 + \mu_2$ enters the HJB), but they determine the P&L attribution between the spot and perp legs.

INVENTORY RISK BRAKE. The feedback coefficient $\gamma_N = [\mu(1 - \alpha_2) + 2\alpha_1]/(2\lambda)$ governs how the optimal rate responds to the existing position size. For sufficiently large inventory penalty φ , the term $2\alpha_1$ dominates and $\gamma_N < 0$: the optimal policy is *self-limiting*, reducing the accumulation rate as N grows. This reflects the quadratic risk penalty φN^2 , which makes the marginal value of adding to the position decline as the book grows. In the closed-loop system (28), this negative feedback ensures that N_t^γ remains bounded in expectation.

BASLINE ACCUMULATION DRIFT. The constant feedback coefficient $\gamma_0 = (\alpha_3 - \mu\alpha_5)/(2\lambda)$ encodes a baseline rate of accumulation driven by the staking yield r and the long-run basis mean m . Even when $D^\gamma = 0$ and $N^\gamma = 0$, the protocol optimally accumulates at rate $\gamma_0 > 0$ provided $r > 0$ and $m > 0$, reflecting the unconditional attractiveness of the delta-neutral carry trade. Starting from zero, γ_0 governs how quickly the protocol initiates its position.

PARAMETER SENSITIVITIES. Four key parameters modulate the aggressiveness of the strategy:

- A higher *discount rate* ρ makes the protocol more impatient, reducing the value of future carry and leading to smaller γ_D and γ_0 and a lower steady-state position.
- A higher *inventory penalty* φ tightens the risk constraint, driving α_1 more negative and reducing the equilibrium position size. In the limit $\varphi \rightarrow \infty$ the optimal position converges to zero; as $\varphi \rightarrow 0$ the protocol accumulates without bound (regulated only by the permanent impact mechanism).
- A higher *permanent impact* $\mu = \mu_1 + \mu_2$ has two opposing effects: it raises the MTM gain $\mu\gamma N^\gamma$ in the running reward, but also compresses the basis more aggressively via (2), destroying future funding income faster. The optimal policy internalizes this by moderating γ_D as μ grows. The individual values of μ_1 and μ_2 do not affect the optimal control (only their sum μ enters the HJB), but they determine the P&L attribution between the spot and perpetual legs.
- A higher *temporary impact* $\lambda = \lambda_1 + \lambda_2$ penalizes fast trading directly. Since all three feedback coefficients scale as $1/\lambda$, a larger λ uniformly slows the optimal accumulation, spreading position changes over a longer horizon. As $\lambda \rightarrow 0$, temporary friction vanishes and the only brakes on accumulation are the permanent impact μ and the inventory penalty φ . The coefficients λ_1 and λ_2 individually determine the slippage attribution between the spot and perpetual legs.

NEGATIVE FUNDING. When the basis turns negative ($D^\gamma < 0$), the carry term $(q + \kappa)D^\gamma N^\gamma$ becomes negative for $N^\gamma > 0$, imposing a carry cost. The feedback coefficient $\gamma_D > 0$ then drives $\gamma^* = \gamma_N N^\gamma + \gamma_D D^\gamma + \gamma_0$ downward (since $D^\gamma < 0$), causing the protocol to reduce its position. The speed of de-risking is controlled by $|\gamma_D|$: a protocol with high φ (large $|\gamma_N|$ and negative γ_N) and large μ will unwind rapidly, while a more patient protocol with small ρ may hold through transient negative-funding episodes and await mean reversion. This behavior is consistent with the observed Ethena protocol response during periods of market stress.

MEAN-VARIANCE INTERPRETATION. The quadratic inventory penalty $\varphi \cdot (N^\gamma)^2$ approximates a mean-variance objective: setting $\varphi = \frac{1}{2}c^2\eta$ for risk-aversion coefficient η makes $\varphi \cdot (N^\gamma)^2 = \frac{1}{2}c^2\eta \cdot (N^\gamma)^2$ proportional to the instantaneous variance of the wealth increment $-cN^\gamma dW$ in (4). Under this identification, the optimal policy (14) is the continuous-time analog of the mean-variance efficient frontier applied to the carry-versus-basis-risk trade-off.

4 FINITE-HORIZON STOCHASTIC CONTROL PROBLEM

4.1 VALUE FUNCTION AND HJB EQUATION

We now consider a finite-horizon variant in which the protocol operates over a fixed interval $[0, T]$ and seeks to maximize expected total wealth accumulated by time T . At the terminal date the protocol must close its remaining position N_T^γ by liquidating both legs simultaneously. Liquidating N_T^γ units at once incurs a lump-sum cost modeled as $\frac{1}{2}\lambda_T(N_T^\gamma)^2$, where $\lambda_T > 0$ is a terminal liquidation-cost coefficient reflecting the price impact of closing a large position quickly. The finite-horizon value function is

$$\begin{aligned} V^T(t, d, n) &:= \sup_{\gamma \in \mathcal{G}} \mathbb{E} \left[\int_t^T \left(dX_s^\gamma - \varphi \cdot (N_s^\gamma)^2 ds \right) - \frac{1}{2}\lambda_T \cdot (N_T^\gamma)^2 \mid D_t^\gamma = d, N_t^\gamma = n \right] \\ &= \sup_{\gamma \in \mathcal{G}} \mathbb{E} \left[\int_t^T \left([(q + \kappa)D_s^\gamma - \kappa m + r + \mu \gamma_s] N_s^\gamma - \lambda \gamma_s^2 - \varphi (N_s^\gamma)^2 \right) ds - \frac{1}{2}\lambda_T \cdot (N_T^\gamma)^2 \mid D_t^\gamma = d, N_t^\gamma = n \right], \end{aligned} \quad (30)$$

where we set the discount rate to zero ($\rho = 0$), as is natural when the objective is total accumulated wealth rather than a discounted present value. The terminal condition is

$$V^T(T, d, n) = -\frac{1}{2}\lambda_T n^2. \quad (31)$$

REMARK 5 (TERMINAL PENALTY). The term $-\frac{1}{2}\lambda_T n^2$ penalizes the protocol for holding a residual position at expiry. It may be interpreted as the execution cost of an instantaneous block liquidation of n units at terminal impact cost λ_T , or as a bequest function that assigns negative value to unhedged residual inventory. As $\lambda_T \rightarrow \infty$, the protocol is forced to unwind fully before T ; as $\lambda_T \rightarrow 0$, the terminal inventory is left unconstrained.

By the principle of dynamic programming, $V^T : [0, T] \times \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfies the HJB PDE

$$-\partial_t V^T = \sup_{\gamma \in \mathbb{R}} \left\{ [(q + \kappa)d - \kappa m + r + \mu \gamma] n - \lambda \gamma^2 - \varphi n^2 + [-\kappa(d - m) - \mu \gamma] \partial_d V^T + \gamma \partial_n V^T + \frac{c^2}{2} \partial_{dd} V^T \right\},$$

with terminal condition (31). The first-order condition for γ is identical in structure to (7):

$$\mu n - 2\lambda \gamma - \mu \partial_d V^T + \partial_n V^T = 0 \implies \gamma^*(t, d, n) = \frac{\mu n - \mu \partial_d V^T + \partial_n V^T}{2\lambda}, \quad (32)$$

and substituting back gives the reduced HJB:

$$-\partial_t V^T = [(q + \kappa)d - \kappa m + r] n - \varphi n^2 - \kappa(d - m) \partial_d V^T + \frac{(\mu n - \mu \partial_d V^T + \partial_n V^T)^2}{4\lambda} + \frac{c^2}{2} \partial_{dd} V^T. \quad (33)$$

4.2 EXPLICIT SOLUTION

We seek a solution of the form

$$V^T(t, d, n) = A(t) n^2 + B(t) n d + C(t) n + E(t) d^2 + F(t) d + G(t), \quad (34)$$

with time-dependent coefficients satisfying the terminal conditions

$$A(T) = -\frac{1}{2}\lambda_T, \quad B(T) = E(T) = C(T) = F(T) = G(T) = 0. \quad (35)$$

The partial derivatives are

$$\partial_d V^T = Bn + 2Ed + F, \quad \partial_n V^T = 2An + Bd + C, \quad \partial_{dd} V^T = 2E, \quad (36)$$

and the combined numerator of (32) is

$$\Phi^T(t, d, n) := \mu n - \mu \partial_d V^T + \partial_n V^T = \underbrace{(\mu(1-B) + 2A)}_{=: 2\lambda\Gamma_N(t)} n + \underbrace{(B - 2\mu E)}_{=: 2\lambda\Gamma_D(t)} d + \underbrace{(C - \mu F)}_{=: 2\lambda\Gamma_0(t)},$$

where we have defined

$$\Gamma_N(t) := \frac{\mu(1-B(t)) + 2A(t)}{2\lambda}, \quad \Gamma_D(t) := \frac{B(t) - 2\mu E(t)}{2\lambda}, \quad \Gamma_0(t) := \frac{C(t) - \mu F(t)}{2\lambda}. \quad (37)$$

If the value function is of the form (34), then the optimal control at time t is given by

$$\gamma^*(t, D_t^\gamma, N_t^\gamma) = \Gamma_N(t) N_t^\gamma + \Gamma_D(t) D_t^\gamma + \Gamma_0(t). \quad (38)$$

We are now in a position to present our main result for the finite-horizon stochastic control problem.

PROPOSITION 6 (FINITE-HORIZON EXPLICIT SOLUTION). *Under the parameter conditions of Proposition 2 with $\rho = 0$, the finite-horizon value function (30) is given by the quadratic (34) and the optimal control is the time-dependent linear feedback (38), where the coefficient functions $A, B, C, E, F, G : [0, T] \rightarrow \mathbb{R}$ satisfy the Riccati ODEs (39)–(40), the linear ODEs (41)–(42) (equivalently, the scalar ODE (43) with explicit integral solution (44)), and the quadrature (45), all solved backward from the terminal conditions (35).*

Proof. Substituting the ansatz (34) and its partial derivatives (36) into the reduced HJB (33) and collecting by monomial in (d, n) yields the following system.

Coefficient of n^2 — Riccati ODE for A:

$$\dot{A}(t) = \varphi - \lambda \Gamma_N(t)^2 = \varphi - \frac{[\mu(1-B) + 2A]^2}{4\lambda}, \quad A(T) = -\frac{1}{2}\lambda T. \quad (39)$$

Coefficient of d^2 — Riccati ODE for E:

$$\dot{E}(t) = (2\kappa - c^2) E - \lambda \Gamma_D(t)^2 = (2\kappa - c^2) E - \frac{(B - 2\mu E)^2}{4\lambda}, \quad E(T) = 0.$$

Coefficient of nd — Riccati ODE for B:

$$\dot{B}(t) = -(q + \kappa) + \kappa B - \frac{[\mu(1-B) + 2A](B - 2\mu E)}{2\lambda}, \quad B(T) = 0. \quad (40)$$

Equations (39)–(40) form a closed Riccati system for (A, E, B) , to be solved first (backward from T).

Coefficient of n — linear ODE for C: Once (A, B, E) are known, collecting the n -coefficient gives

$$\dot{C}(t) = -(r - \kappa m) - \kappa m B - \Gamma_N(t)(C - \mu F), \quad C(T) = 0. \quad (41)$$

Coefficient of d — linear ODE for F:

$$\dot{F}(t) = \kappa F - 2\kappa m E - \Gamma_D(t)(C - \mu F), \quad F(T) = 0. \quad (42)$$

Setting $\mathcal{C}(t) := C(t) - \mu F(t) = 2\lambda\Gamma_0(t)$ and subtracting μ times (42) from (41):

$$\dot{\mathcal{C}}(t) = -(r - \kappa m(1 - B + 2\mu E)) - (\Gamma_N(t) - \mu\Gamma_D(t) + \mu\kappa)\mathcal{C}(t), \quad \mathcal{C}(T) = 0. \quad (43)$$

This is a scalar linear ODE with explicit integral solution

$$\mathcal{C}(t) = \int_t^T h(s) \exp\left(\int_t^s [\Gamma_N(u) - \mu\Gamma_D(u) + \mu\kappa] du\right) ds, \quad (44)$$

where $h(s) := r - \kappa m(1 - B(s) + 2\mu E(s))$. Once $\mathcal{C}$ is known, F satisfies the scalar linear ODE (42) and $C = \mathcal{C} + \mu F$.

Constant term:

$$\dot{G}(t) = -\lambda\Gamma_0(t)^2 - c^2E(t) - \kappa mF(t), \quad G(T) = 0, \quad (45)$$

which is obtained by direct integration once (E, F, Γ_0) are known. $\square$

REMARK 7 (HIERARCHICAL STRUCTURE OF THE ODE SYSTEM). The backward ODE system has a hierarchical cascade mirroring the algebraic cascade in the infinite-horizon proof. The Riccati block (A, E, B) in (39)–(40) is independent of (C, F, G) and is solved first. The linear block (C, F) is then driven by the already-solved Riccati block; it reduces to the single scalar ODE (43) with explicit integral solution (44). Finally, G follows by quadrature from (45).

REMARK 8 (SPECIAL CASE: NO PERMANENT IMPACT: $\mu = 0$). When $\mu = 0$ the system decouples. The ODE for B reduces to $\dot{B} = -(q + \kappa) + \kappa B$ with $B(T) = 0$, giving

$$B(t) = \frac{q + \kappa}{\kappa} (1 - e^{-\kappa(T-t)}),$$

and the ODE for A reduces to the scalar Riccati equation $\dot{A} = \varphi - A^2/\lambda$ with $A(T) = -\frac{1}{2}\lambda_T$, which has the explicit solution

$$A(t) = -\sqrt{\lambda\varphi} \frac{\sqrt{\lambda\varphi} + \lambda_T \tanh\left(\sqrt{\varphi/\lambda}(T-t)\right)}{\lambda_T + \sqrt{\lambda\varphi} \tanh\left(\sqrt{\varphi/\lambda}(T-t)\right)}.$$

4.3 ECONOMIC IMPLICATIONS

The finite-horizon solution shares several qualitative features with the infinite-horizon solution of Section 3, but the time-dependence of the feedback coefficients $\Gamma_N(t)$, $\Gamma_D(t)$, $\Gamma_0(t)$ introduces important new phenomena.

LIQUIDATION URGENCY NEAR EXPIRY. As $t \rightarrow T$, the terminal condition $A(T) = -\frac{1}{2}\lambda_T < 0$ pulls $\Gamma_N(t)$ sharply negative. From (37), $\Gamma_N(t) \approx -\lambda_T/\lambda < 0$ for t near T , which drives $\gamma^* \approx (-\lambda_T/\lambda)N_t^\gamma$ — a rate of unwinding proportional to the current position size. This liquidation urgency is entirely absent from the infinite-horizon policy and reflects the finite deadline: the protocol must close its position regardless of the current basis level.

BASIS CHASING EARLY, DELEVERAGING LATE. Far from expiry ($T - t$ large), the time-dependent feedback coefficients converge to their infinite-horizon counterparts: $\Gamma_D(t) \rightarrow \gamma_D > 0$ and $\Gamma_N(t) \rightarrow \gamma_N$. The protocol therefore pursues the same basis-chasing, inventory-risk-braking policy as in the infinite-horizon case when time to expiry is long. As $t \rightarrow T$, the basis-chasing feedback coefficient $\Gamma_D(t)$ diminishes because the remaining horizon over which future funding can be collected shrinks, making basis-driven accumulation less attractive relative to the cost of the terminal liquidation penalty.

INTERACTION BETWEEN TERMINAL COST AND INVENTORY PENALTY. The terminal cost λ_T and the running inventory penalty φ both discourage large positions, but through different channels. The running penalty φ operates continuously, slowing accumulation throughout $[0, T]$. The terminal cost λ_T operates only at T but propagates backward through the Riccati ODE (39), shaping the entire time profile of $\Gamma_N(t)$. A large λ_T (high liquidation cost) induces early de-risking, effectively front-loading the liquidation and spreading it over $[0, T]$ rather than concentrating it at T .

CONVERGENCE TO THE INFINITE-HORIZON SOLUTION. With a discount rate $\rho > 0$ reintroduced, the time-dependent coefficients $(A(t), B(t), E(t), \dots)$ converge as $T \rightarrow \infty$ to the stationary values $(\alpha_1, \alpha_2, \alpha_4, \dots)$ of Proposition 2, and the feedback coefficients $(\Gamma_N(t), \Gamma_D(t), \Gamma_0(t))$ converge to the stationary feedback coefficients $(\gamma_N, \gamma_D, \gamma_0)$. The finite-horizon solution thus provides a natural generalization that nests the infinite-horizon result as a limiting case.

CLOSED-LOOP DYNAMICS. Under the optimal control (38), the state processes satisfy the two-dimensional linear SDE

$$dD_t^\gamma = [-(\kappa + \mu\Gamma_D(t))D_t^\gamma - \mu\Gamma_N(t)N_t^\gamma + \kappa m - \mu\Gamma_0(t)]dt + c dW_t, \quad (46)$$

$$dN_t^\gamma = [\Gamma_N(t)N_t^\gamma + \Gamma_D(t)D_t^\gamma + \Gamma_0(t)]dt, \quad (47)$$

whose solution is Gaussian for all $t \in [0, T]$. The time-varying drift matrix

$$M(t) := \begin{pmatrix} -(\kappa + \mu\Gamma_D(t)) & -\mu\Gamma_N(t) \\ \Gamma_D(t) & \Gamma_N(t) \end{pmatrix}$$

has the same qualitative structure as the infinite-horizon matrix (29), with time-independent feedback coefficients replaced by time-dependent ones that converge to the infinite-horizon values as $T - t \rightarrow \infty$.

5 NUMERICAL EXPERIMENTS

In this section, we illustrate the optimal Ethena strategy by comparing the time-dependent finite-horizon policy with the stationary infinite-horizon policy. To highlight the qualitative differences between these two regimes, we use the following model parameters:

$$\begin{aligned} \rho &= 0.05, & \kappa &= 2.0, & m &= 0.04, & \mu &= 0.3, & \lambda &= 0.1, & r &= 0.04, & q &= 4.0, \\ c &= 0.1, & \varphi &= 0.5, & \lambda_T &= 4.0, & T &= 1.0, & D_0^\gamma &= m, & N_0^\gamma &= 0.0, & X_0^\gamma &= 0.0. \end{aligned}$$

In Figure 1, we plot $\Gamma_0(t)$, $\Gamma_N(t)$ and $\Gamma_D(t)$ as a function of t (solid red lines). For comparison, we also plot the constants γ_0 , γ_N and γ_D (dashed blue lines). Several key behaviors emerge:

- *Convergence to Stationarity:* For times t far from the terminal date T , the finite-horizon coefficients $(\Gamma_0(t), \Gamma_N(t), \Gamma_D(t))$ are nearly identical to the infinite-horizon values $(\gamma_0, \gamma_N, \gamma_D)$. This suggests that for a protocol with a long operational runway, the stationary policy provides a highly accurate and computationally simpler approximation of the optimal strategy.
- *Liquidation Urgency:* As $t \rightarrow T$, we observe a sharp divergence in the coefficients. Most notably, $\Gamma_N(t)$ becomes significantly more negative. This reflects a “liquidation urgency”; the high terminal cost λ_T incentivizes the protocol to begin unwinding its position well before the deadline to avoid a prohibitive lump-sum execution cost at time T .

- *Basis Sensitivity:* The coefficient $\Gamma_D(t)$, which governs the protocol’s response to the basis (funding-rate chasing), remains relatively stable until the final stages of the horizon. This indicates that the protocol continues to harvest carry from basis fluctuations even late in its lifecycle, though this response is eventually dampened by the overarching requirement to reach a zero position.

In Figure 2, we plot one sample path of D_t^γ , N_t^γ and X_t^γ for both the finite horizon case (solid red lines) and infinite horizon case (dashed blue lines). We note the following:

- *Position Smoothing:* Under the finite-horizon policy, the position N_t^γ (red) begins to deviate from the infinite-horizon path (blue) as t approaches T . While the infinite-horizon strategy—lacking a deadline—continues to maintain a large short position to harvest carry, the finite-horizon strategy yields a smoother transition toward zero.
- *Wealth Trade-offs:* The cumulative wealth X_t^γ highlights the internalized cost of terminal liquidation. The red path demonstrates how the protocol “pays” for its liquidation early through reduced accumulation rates. This proactive unwinding ensures the protocol does not face a catastrophic wealth drop at time T due to the terminal penalty $\frac{1}{2}\lambda_T(N_T^\gamma)^2$.
- *Basis Interaction:* The basis paths D_t^γ remain closely coupled for much of the duration. However, as $t \rightarrow T$, the finite-horizon path experiences less self-induced compression. By reducing its trading rate near the horizon, the protocol allows the basis to mean-revert more naturally toward m without the persistent downward pressure exerted by the protocol’s own short-selling.

Mathematica code that was used to create the plots is provided in Appendix A.

6 CONCLUSION AND FUTURE RESEARCH

In this paper, we have developed a rigorous stochastic control framework for the optimal management of the Ethena protocol’s delta-neutral carry strategy. By modeling the simultaneous execution of staked asset purchases and perpetual futures shorting, we derived optimal feedback policies for both infinite and finite horizons. Our results demonstrate that the optimal strategy is characterized by a balance between three competing forces: the desire to harvest staking rewards and funding income, the need to minimize permanent and temporary market impact, and the mitigation of inventory risk.

There are several promising directions for future research:

- *Endogenous Inflows and Outflows:* Perhaps the most significant extension involves modeling the Ethena position N_t not as a pure control variable, but as a process driven by user deposits and withdrawals. In the current model, the controller chooses the rate of change γ_t to maximize wealth. In reality, the protocol must execute trades in response to minting and redeeming demand for the stablecoin. Formulating this as a tracking problem—where the protocol seeks to maintain a delta-neutral hedge while minimizing the impact costs associated with stochastic AUM fluctuations—would be of great practical interest.
- *Stochastic Parameters and Regime Switching:* While we modeled the basis as a mean-reverting process, the staking rewards r and the funding sensitivity q were treated as constants. Future work could incorporate stochastic volatility or regime-switching dynamics to capture “funding rate spikes” or periods of extreme market stress where the basis deviates significantly from historical norms.

- *Multi-Asset Portfolio Optimization:* Ethena has expanded to include assets beyond Ethereum, such as Bitcoin and Solana. Extending this framework to a multi-asset setting would allow for the study of optimal collateral diversification and the cross-impact effects of simultaneous hedging across multiple perpetual markets.
- *Counterparty and Protocol Risk:* Integrating a “jump-to-default” or a credit risk component to account for the potential failure of centralized exchanges or the de-pegging of the staked asset would provide a more comprehensive view of the protocol’s risk-adjusted returns.

A Wolfram Mathematica © CODE

In this Appendix, we provide the *Wolfram Mathematica* © code that was used to produce the plots in Section 5.

```

1  (* 1. MODEL PARAMETERS *)
2  (* These parameters are chosen to satisfy the stability condition for the infinite
   horizon *)
3  rho = 0.05;      (* Discount rate *)
4  kappa = 2.0;     (* Basis mean-reversion speed (x in the model) *)
5  m = 0.04;        (* Long-run basis level *)
6  mu = 0.3;        (* Total permanent impact *)
7  lambda = 0.1;    (* Total temporary impact *)
8  r = 0.04;        (* Staking yield *)
9  q = 4.0;         (* Funding rate factor *)
10 c = 0.1;         (* Basis volatility *)
11 phi = 0.5;       (* Inventory risk penalty *)
12 lambdaT = 4.0;   (* Terminal liquidation cost *)
13 Tmax = 1.0;     (* Time horizon *)
14
15 xStar = rho + 2*kappa - c^2;
16
17 (* 2. INFINITE HORIZON SOLUTION *)
18 a1Inf[a2_] := (lambda*rho - mu*(1 - a2) - Sqrt[(lambda*rho - mu*(1 - a2))^2 + 4*lambda*
   phi - mu^2*(1 - a2)^2])/2;
19 a4Inf[a2_] := (mu*a2 + lambda*xStar - Sqrt[lambda*xStar*(lambda*xStar + 2*mu*a2)])/(2*mu
   ^2);
20
21 valA2 = a2 /. FindRoot[(rho + kappa)*a2 == (q + kappa) + ((mu*(1 - a2) + 2*a1Inf[a2])*(a2
   - 2*mu*a4Inf[a2]))/(2*lambda), {a2, (q + kappa)/(rho + kappa)}];
22
23 infA1 = a1Inf[valA2];
24 infA4 = a4Inf[valA2];
25 infA2 = valA2;
26
27 gammaNInf = (mu*(1 - infA2) + 2*infA1)/(2*lambda);
28 gammaDInf = (infA2 - 2*mu*infA4)/(2*lambda);

```

```

29 cInf = (r + kappa*m*(infA2 - 2*mu*infA4))/(rho - gammaNInf + mu*gammaDInf);
30 gamma0Inf = cInf/(2*lambda);
31
32 (* 3. FINITE HORIZON SOLUTION *)
33 finiteODE = NDSolve[{
34   a'[t] == phi - (mu*(1 - b[t]) + 2*a[t])^2/(4*lambda),
35   e'[t] == (2*kappa - c^2)*e[t] - (b[t] - 2*mu*e[t])^2/(4*lambda),
36   b'[t] == -(q + kappa) + kappa*b[t] - (mu*(1 - b[t]) + 2*a[t])*(b[t] - 2*mu*e[t])/(2*
      lambda),
37   cv'[t] == -(r + kappa*m*(1 - b[t] + 2*mu*e[t])) - ((mu*(1 - b[t]) + 2*a[t])/(2*lambda
      ) - mu*(b[t] - 2*mu*e[t])/(2*lambda) + mu*kappa)*cv[t],
38   a[Tmax] == -lambdaT/2, b[Tmax] == 0, e[Tmax] == 0, cv[Tmax] == 0
39 }, {a, b, e, cv}, {t, 0, Tmax}][[1]];
40
41 GammaN[t_] := (mu*(1 - b[t]) + 2*a[t])/(2*lambda) /. finiteODE;
42 GammaD[t_] := (b[t] - 2*mu*e[t])/(2*lambda) /. finiteODE;
43 Gamma0[t_] := (cv[t])/(2*lambda) /. finiteODE;
44
45 (* 4. PLOTS OF FEEDBACK CONTROL COEFFICIENTS *)
46 Print["Comparison of constant feedback coefficient gamma_0 (Blue, Dashed) and time-
      dependent feedback coefficient Gamma_0(t) (Red, Solid)"];
47 Plot[{gamma0Inf, Gamma0[t]}, {t, 0, Tmax}, PlotStyle -> {{Blue, Dashed}, {Red}}, Frame ->
      True, PlotRange -> Full]
48
49 Print["Comparison of constant feedback coefficient gamma_N (Blue, Dashed) and time-
      dependent feedback coefficient Gamma_N(t) (Red, Solid)"];
50 Plot[{gammaNInf, GammaN[t]}, {t, 0, Tmax}, PlotStyle -> {{Blue, Dashed}, {Red}}, Frame ->
      True, PlotRange -> Full]
51
52 Print["Comparison of constant feedback coefficient gamma_D (Blue, Dashed) and time-
      dependent feedback coefficient Gamma_D(t) (Red, Solid)"];
53 Plot[{gammaDInf, GammaD[t]}, {t, 0, Tmax}, PlotStyle -> {{Blue, Dashed}, {Red}}, Frame ->
      True, PlotRange -> Full]
54
55 (* 5. SIMULATION *)
56 dt = 0.001; steps = Round[Tmax/dt];
57 times = Table[i*dt, {i, 0, steps}];
58 dW = RandomVariate[NormalDistribution[0, Sqrt[dt]], steps];
59
60 (* Paths: [[1]] is Infinite Horizon (Blue), [[2]] is Finite Horizon (Red) *)
61 pathD = {{m}, {m}}; pathN = {{0.}, {0.}}; pathX = {{0.}, {0.}};
62
63 Do[
64   currT = (i - 1)*dt;
65

```



```

66      (* Controls *)
67      gInf = gammaNInf*Last[pathN[[1]]] + gammaDInf*Last[pathD[[1]]] + gamma0Inf;
68      gFin = GammaN[currT]*Last[pathN[[2]]] + GammaD[currT]*Last[pathD[[2]]] + Gamma0[currT]
        ];
69
70      (* Dynamics for Infinite Horizon *)
71      AppendTo[pathD[[1]], Last[pathD[[1]]] + (-kappa*(Last[pathD[[1]]] - m) - mu*gInf)*dt
        + c*dW[[i]]];
72      AppendTo[pathN[[1]], Last[pathN[[1]]] + gInf*dt];
73      AppendTo[pathX[[1]], Last[pathX[[1]]] + ((q + kappa)*pathD[[1]][[-2]] - kappa*m + r +
        mu*gInf)*pathN[[1]][[-2]]*dt - lambda*gInf^2*dt - c*pathN[[1]][[-2]]*dW[[i]]];
74
75      (* Dynamics for Finite Horizon *)
76      AppendTo[pathD[[2]], Last[pathD[[2]]] + (-kappa*(Last[pathD[[2]]] - m) - mu*gFin)*dt
        + c*dW[[i]]];
77      AppendTo[pathN[[2]], Last[pathN[[2]]] + gFin*dt];
78      AppendTo[pathX[[2]], Last[pathX[[2]]] + ((q + kappa)*pathD[[2]][[-2]] - kappa*m + r +
        mu*gFin)*pathN[[2]][[-2]]*dt - lambda*gFin^2*dt - c*pathN[[2]][[-2]]*dW[[i]]];
79      , {i, 1, steps}];
80
81      Print["(i) Basis paths D_t: Infinite (Blue, Dashed) vs Finite (Red, Solid)"];
82      ListLinePlot[{Transpose[{times, pathD[[1]]}], Transpose[{times, pathD[[2]]}]}, PlotStyle
        -> {{Blue, Dashed}, {Red}}, Frame -> True, PlotRange -> Full]
83
84      Print["(ii) Position paths N_t: Infinite (Blue, Dashed) vs Finite (Red, Solid)"];
85      ListLinePlot[{Transpose[{times, pathN[[1]]}], Transpose[{times, pathN[[2]]}]}, PlotStyle
        -> {{Blue, Dashed}, {Red}}, Frame -> True, PlotRange -> Full]
86
87      Print["(iii) Net Wealth paths X_t: Infinite (Blue, Dashed) vs Finite (Red, Solid)"];
88      ListLinePlot[{Transpose[{times, pathX[[1]]}], Transpose[{times, pathX[[2]]}]}, PlotStyle
        -> {{Blue, Dashed}, {Red}}, Frame -> True, PlotRange -> Full]

```

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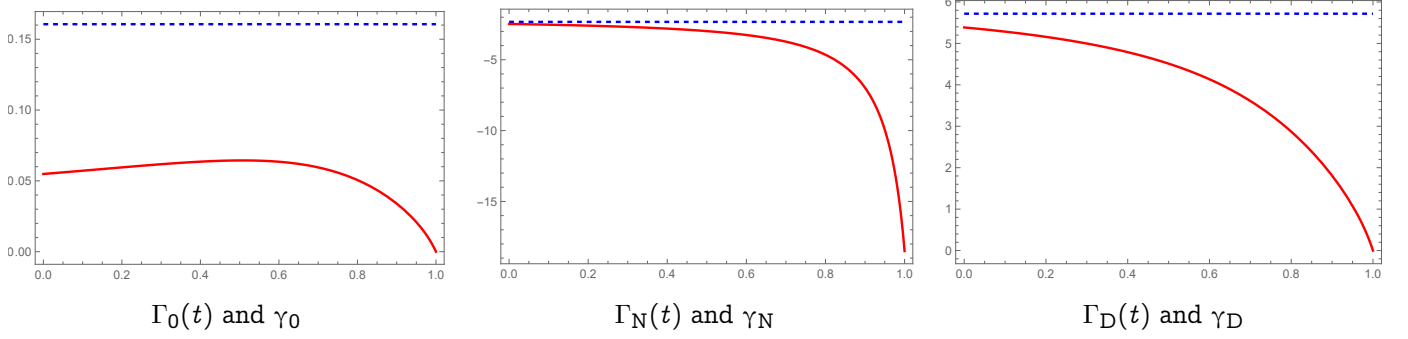


Figure 1: Using the model parameters given in Section 5 we plot $\Gamma_0(t)$, $\Gamma_N(t)$ and $\Gamma_D(t)$, given in (37), as functions of t (red lines). For comparison, we also plot γ_0 , γ_N and γ_D , given in (12) (dashed blue lines).

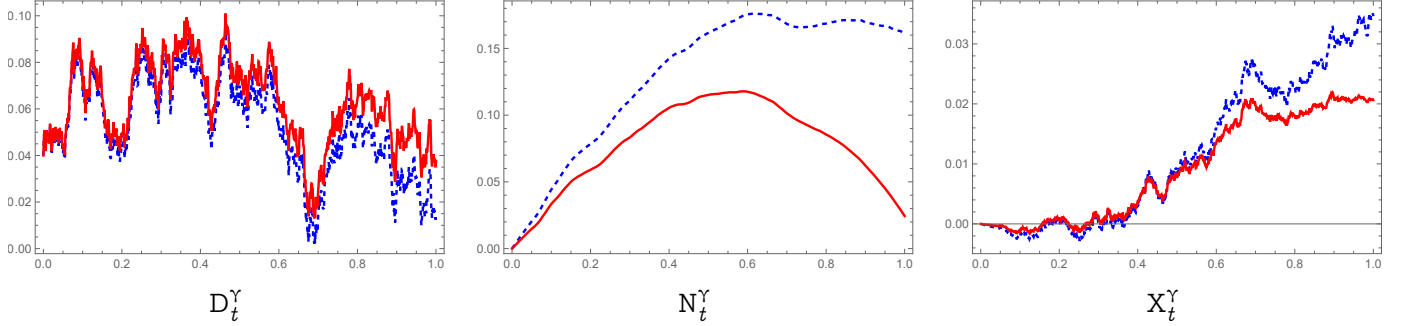


Figure 2: Using the model parameters given in Section 5 we plot one sample path of D_t^Y , N_t^Y and X_t^Y for both the finite horizon case (red lines) and the infinite horizon case (dashed blue lines). Recall that the closed-loop dynamics of D_t^Y and N_t^Y in the infinite horizon case are given by (27) and (28), respectively, while the closed-loop dynamics of D_t^Y and N_t^Y in the finite horizon case are given by (46) and (47), respectively. In both the infinite and finite horizon cases the dynamics of X_t^Y are given by (4) with γ_t given by (14) and (38) in the infinite horizon and finite horizon cases, respectively.